Manufacturing

- Arrived from the Latin word “*manu factus*”, meaning “made by hand”.
- Manufacturing is the process of converting raw materials into products.

Production Processes

Project- or Job-Based Production

- **Project-based production** is one-of-a-kind production in which only one unit is manufactured at a time. This type of production is often used for very large projects or for individual customers. Because the customer's needs and preferences play such a decisive role in the final output, it's essential for the operations manager to maintain open and frequent communication with that customer. The workers involved in this type of production are highly skilled or specialists in their field.

The following are examples of project- or job-based production:

- custom home construction
- haircuts
- yachts

Production Processes

Batch Production

- **Batch production** is a method used to produce similar items in groups, stage by stage. In batch production, the product goes through each stage of the process together before moving on to the next stage. The degree to which workers are involved in this type of production depends on the type of product. It is common for machinery to be used for the actual production and workers participate only at the beginning and end of the process.

Examples of batch production include the following:

- bakeries
- textiles
- furniture

Production Processes

Mass Production

- **Mass production** is used by companies that need to create standardized products in large quantities as economically as possible. Products are mass produced in order to generate the inventory needed to meet high market demand. This type of production usually requires heavy investment in machinery and equipment; workers are generally needed to assemble component parts to make the finished good.

The following goods are mass produced:

- toilet paper
- cell phones
- automobiles

Production Processes

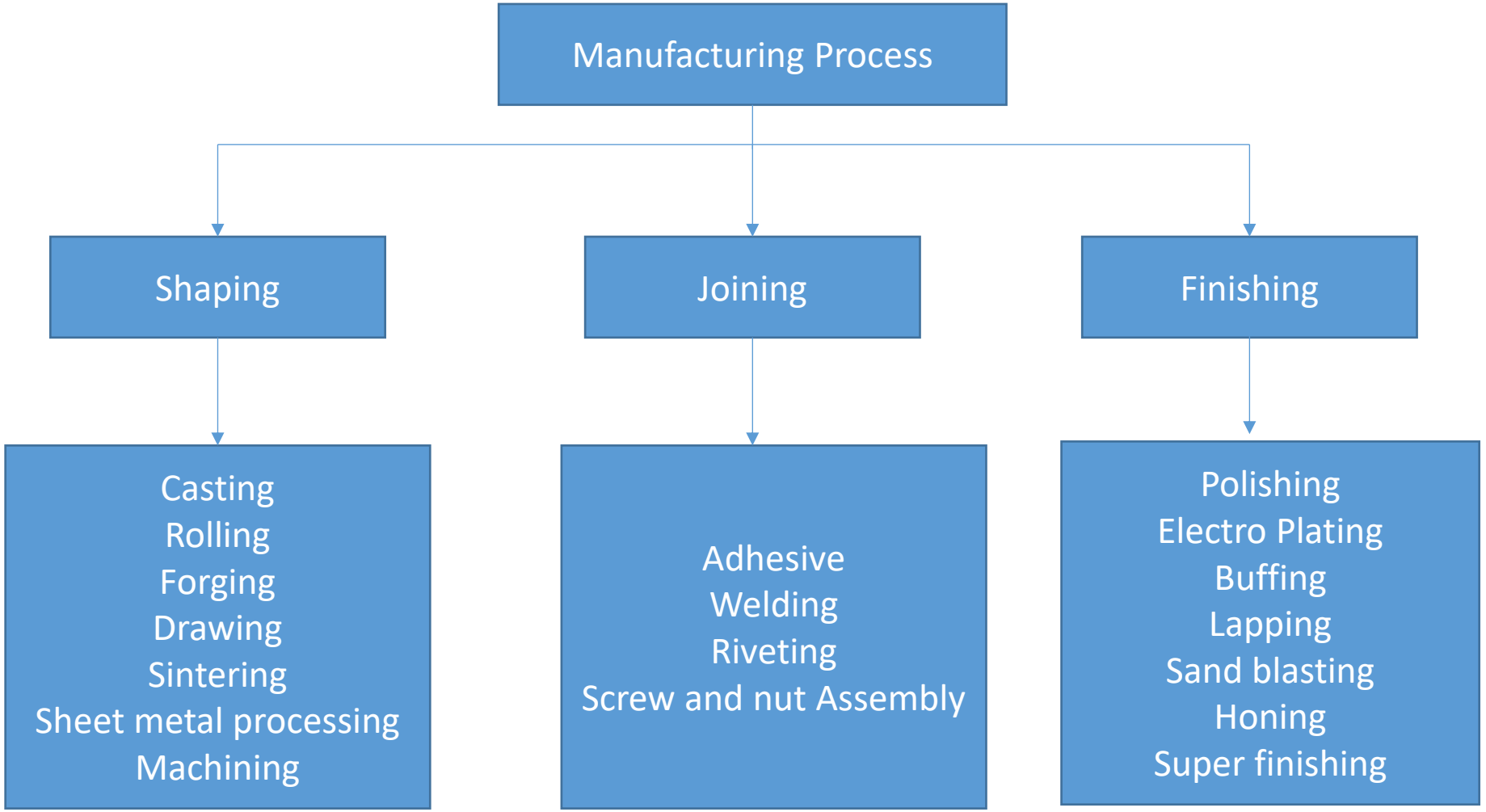
Flow or Continuous Production

- **Flow production**, also known as continuous production, occurs when a process runs twenty-four hours a day. Companies whose products are homogeneous use this production approach to reduce cost and increase efficiency. These systems are highly automated, and workers act as monitors rather than as active participants.

Examples of flow production:

- gas and oil
- steel
- chemicals

Classification of Manufacturing Process



Selection of a process for production

There are hundreds of manufacturing processes. You are likely to already be familiar with the most common, e.g. casting, forming, molding and machining. For any given product, there will be multiple manufacturing processes that you'll need to select from. The process you choose will depend on many factors called the *process selection drivers*. These process selection drivers include the following:

- Quantity of the product
- Cost for tooling, manufacturing machines and equipment
- Time required for processing
- Level of skilled labor required
- Process supervision
- Energy consumption
- Availability of material and cost of material
- Capabilities required to processes material
- Product dimensions and size
- Surface finish required
- Design tolerances
- Waste produced by the process
- Maintenance costs
- Other costs

Recent Developments in Manufacturing

- ***Computer Integrated Manufacturing (CIM)*** combines CAD, CAM, and computer control.
- ***Additive manufacturing*** is another name for 3D printing.
- ***Lean manufacturing*** focuses on eliminating waste and improving efficiency.
- ***Industry 4.0*** represents the digital transformation of manufacturing.
- The use of robots for performing repetitive tasks is known as ***industrial automation***.
- ***Flexible manufacturing system (FMS)*** enables rapid changeover between products.
- ***Smart factories*** use IoT and real-time data analytics.
- ***Sustainable manufacturing*** minimizes environmental impact.
- ***CNC machining*** improves accuracy and repeatability.
- ***Rapid prototyping*** allows quick creation of product models.

Introduction of casting

- A process based on the property of liquid to take up the shape of the vessel containing it.
- A cavity of desired shape is made, contained in a mold.

History of Casting

Ancient process, started 5000 years ago.

Jaivana- 50 tons cannon was built in 17th century in Jaipur.

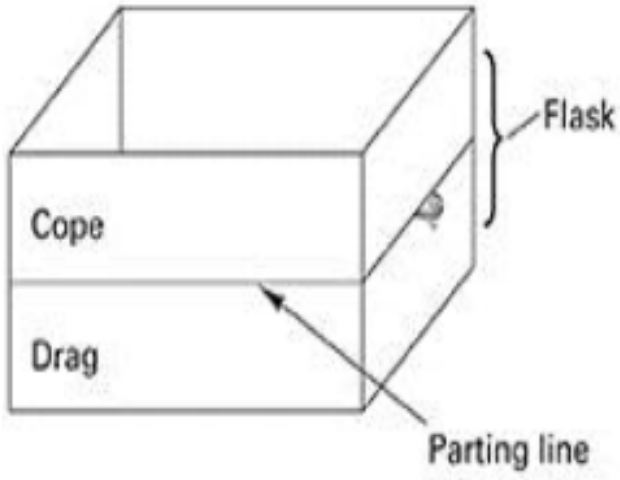
Used for making arrows, coins, knives etc.



World's largest Cannon in Jaipur, made by casting

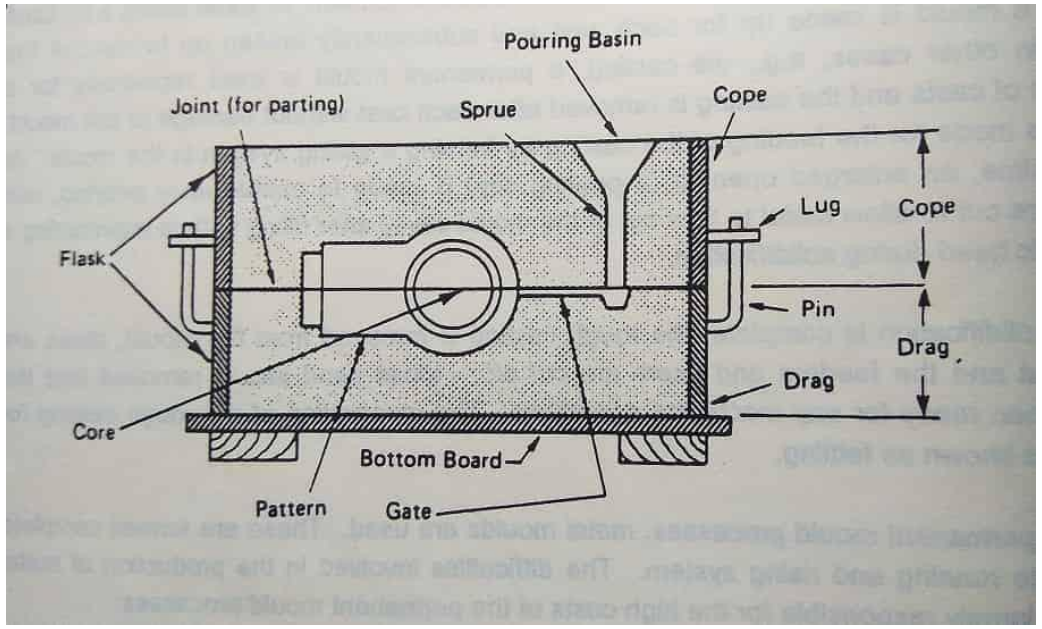
Basic terms used in Casting

1. Pattern: Replica of the desired part
2. Mold- Container with a cavity within. Divided in two halves: Cope and Drag.
3. Gating system- Network of channels that deliver molten metal to the cavity.



Basic steps in Casting

1. Pattern making
2. Mold making
3. Melting of metal and pouring
4. Cooling and solidification of metal
5. Cleaning of casting and inspection



Pattern

- Replica of the desired product
- Has somehow different dimensions than the actual part to be manufactured
- Used to form the mold cavity

Pattern Materials

Requirements:

1. Easily shaped, worked, machined and joined
2. Resistant to wear and corrosion
3. Resistant to chemical action
4. Dimensionally stable
5. Easily available and economical

Pattern Materials

1. Wood:

- Easy availability, low weight and low cost
- Can be easily shaped
- More than 90% castings use wood patterns
- Absorbs moisture. So, distortions and dimensional changes occur
- Relatively lower life, hence suitable for small quantity production

Pattern Materials

2. Metal:

- Used for large quantity production and for closer dimensional tolerances
- Longer life
- Aluminum is mostly used.
- Other metals: cast iron, brass etc.

3. Plastic:

- Low weight, easier formability, smooth surfaces and durability
- Do not absorb moisture. So, dimensionally stable
- Corrosion resistance

Pattern Materials

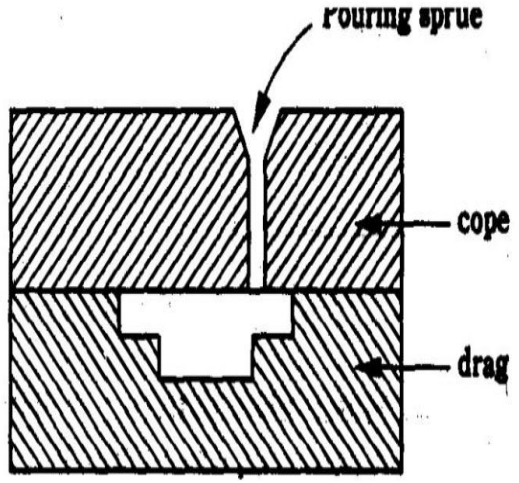
4. Polystyrene:

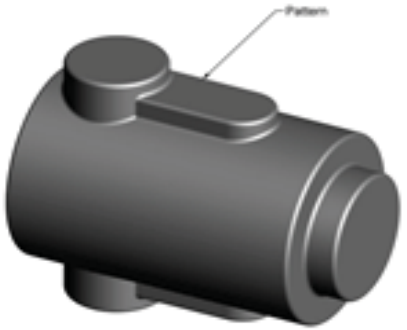
- Changes to gaseous state on heating
- Disposable Patterns. Hence, suitable for single casting.
- When molten metal is poured into cavity, polystyrene transforms to gaseous state.
- Used mostly for small and complicated shaped castings.

Pattern Types

1.Single piece pattern:

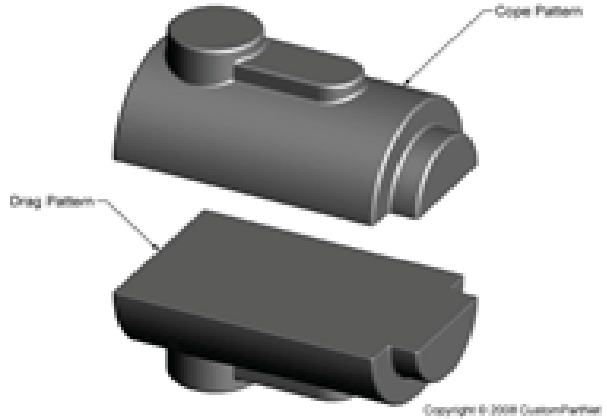
- Used for simple shaped & large castings.
- Pattern and cavity produced by it are completely in the lower flask (i.e. drag)
- Causes difficulty in making the mold.





2.Split pattern:

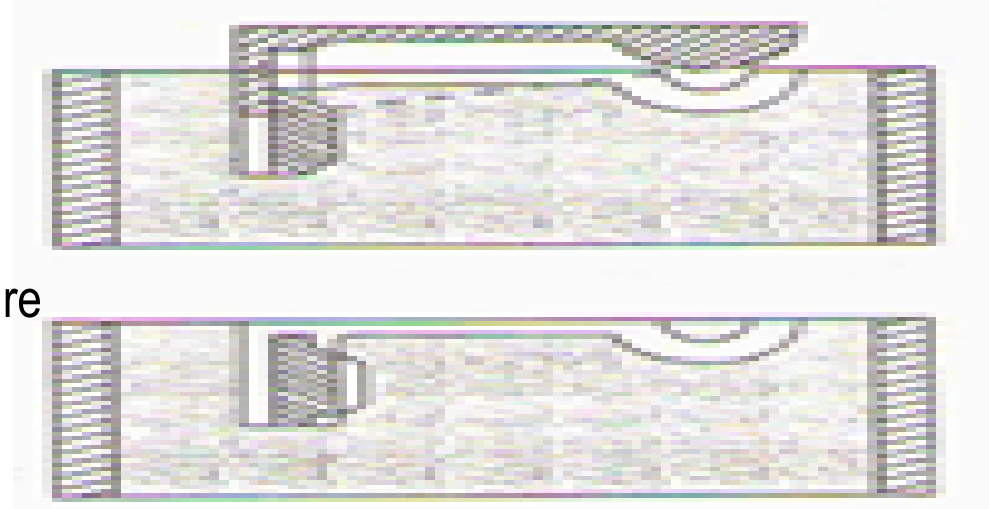
- Split pattern models the part as two separate pieces that meet along the parting line of the mould
- Two parts are aligned by Dowel pin



Pattern Types

3.Loose Piece pattern

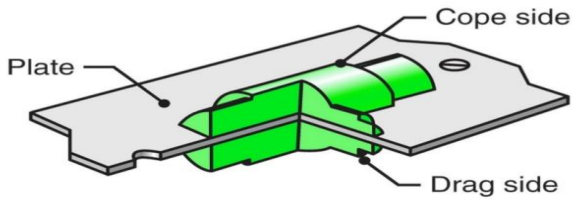
- As pre The name Pattern Contain One or more than one loose piece
- Loose piece is used to make removal of pattern easy from mould box
- loose pieces are attached to main body with the help of dowel pins or wire
- First main pattern is drawn and then carefully loose pieces
- Moulding with this pattern is expensive and require more skill

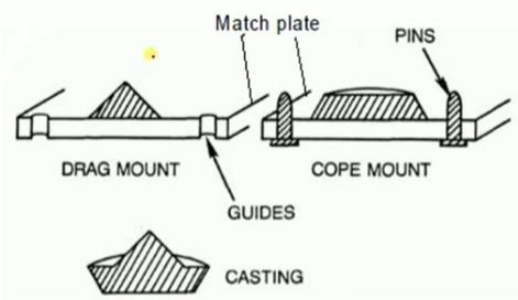


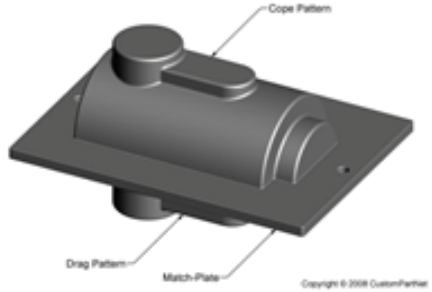
4. Match-plate pattern:

- Similar to a split pattern, except that each half of the pattern is attached to opposite sides of a single plate.
- Match plate confirms the parting line
- Ensures proper alignment of the mould cavities in the cope and drag and the runner system can be included on the match plate.
- Used for larger production quantities.
- Ic Engines piston rings

A typical metal match-plate pattern used in sand casting.





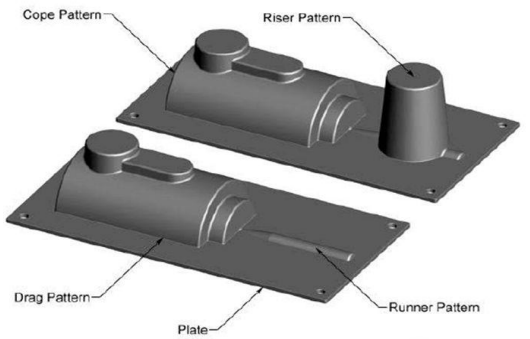


Pattern Types

4. Cope-Drag pattern:

- It is similar to split pattern
- Each half of the pattern is attached to a separate plate and the mould halves are made independently.
- .Pattern is made in two halves and split along parting line
- These two halves are known as cope and drag
- There moulding done independently
- After moulding they are assembled to form complete mould box
- Often desirable for larger castings

Cope and Drag Pattern



Pattern Types

5. Gated pattern:

- In this patters of gate and riser or runner are permanently attached to regular pattern
- They are used to manufacture multiple casting in on time
- Each pattern is connected with common runner .
- Suitable for pouring small castings and for mass production
- It save labour and time



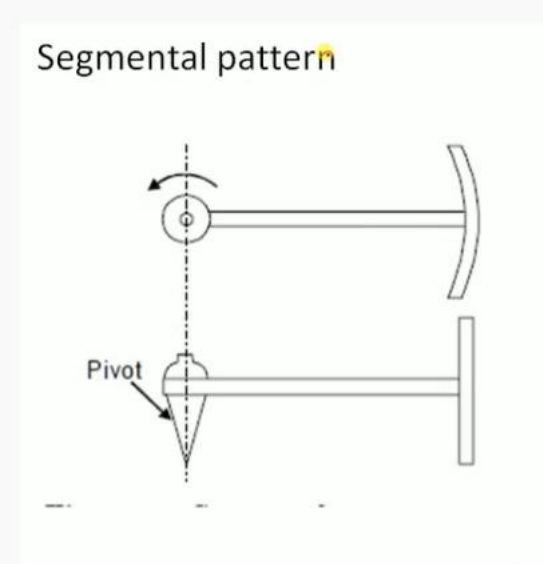
Pattern Types

6. Segmental pattern

Patterns of this type are generally used for circular castings.

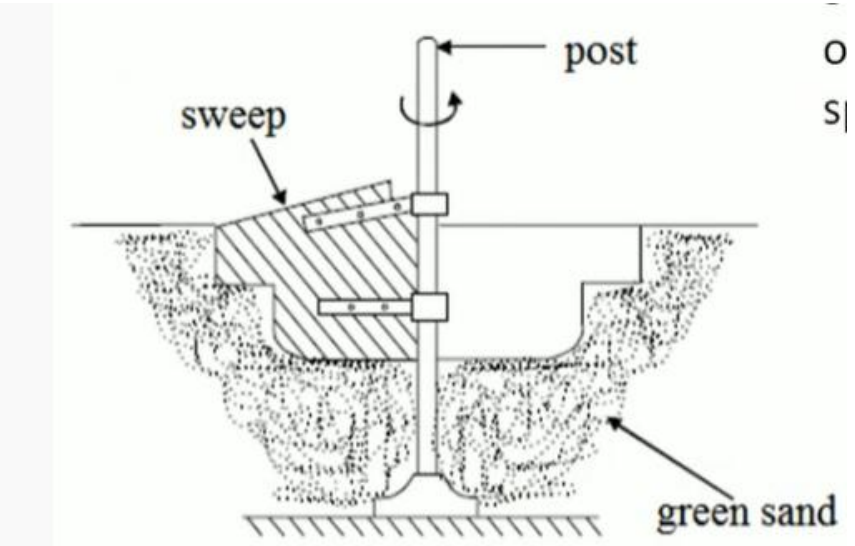
For example wheel rim, gear blank etc. Such patterns are sections of a patterns so arranged as to form a complete mould being moved to form each section of the mould.

The movement of segmental pattern is guided by the use of central pivot.



7. Sweep Patterns

Sweep Patterns are used for forming large circular moulds of symmetric kind by revolving a sweep attached to a spindle.



Pattern Allowances

1. Shrinkage or contraction allowance
2. Draft or taper allowance
3. Machining or finish allowance
4. Distortion or camber allowance
5. Rapping allowance

Pattern Allowances

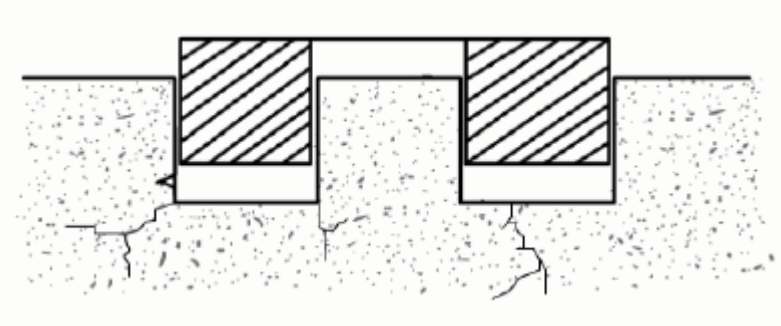
1. Shrinkage Allowance:

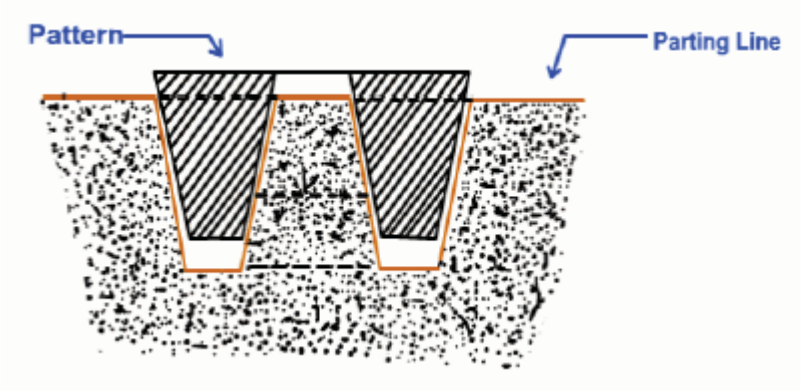
- Shrinkage means contraction of metal on solidification
- All metals shrink after solidification. (Except grey cast iron that expands on solidification).
- It is expressed in mm/m.
- Shrinkage allowance Differs from material to material.
- Positive type of allowance.

Material	Dimension	Shrinkage allowance (inch/ft)
Grey Cast Iron	Up to 2 feet 2 feet to 4 feet over 4 feet	0.125 0.105 0.083
Cast Steel	Up to 2 feet 2 feet to 6 feet over 6 feet	0.251 0.191 0.155
Aluminum	Up to 4 feet 4 feet to 6 feet over 6 feet	0.155 0.143 0.125
Magnesium	Up to 4 feet Over 4 feet	0.173 0.155

Pattern Allowances

2. Draft / Taper Allowance:





- Taper provided by the pattern maker on all vertical surfaces of the pattern so that it can be removed from the sand without tearing away the sides of the sand mould.
- Inner details of the pattern require higher draft than outer surfaces.
- commonly applied draft allowance is 1° to 3°
- due to draught allowance pattern can easily removed from mould box

The amount of draft depends upon-

a) The length of the vertical side of the pattern to be extracted;

b) The intricacy of the pattern;

c) The method of moulding

d) Pattern material.

Pattern Allowances

Pattern material	Height of the given surface (inch)	**Draft angle (External surface)**	**Draft angle (Internal surface)**
Wood	1	3.00	3.00
	1 to 2	1.50	2.50
	2 to 4	1.00	1.50
	4 to 8	0.75	1.00
	8 to 32	0.50	1.00
Metal and plastic	1	1.50	3.00
	1 to 2	1.00	2.00
	2 to 4	0.75	1.00
	4 to 8	0.50	1.00
	8 to 32	0.50	0.75

Pattern Allowances

3. Machining / Finishing Allowance:

- The finish and accuracy in sand casting is generally poor.
- Extra material provided on the casting to enable their machining or finishing to the required size, accuracy and surface finish.
- So it is Added in pattern dimensions.
- allowance is provided only in machining area

Amount of machining allowance depends upon:

a) Method of moulding and casting used
b) Size and shape of casting
c) Casting orientation
d) Metal used in casting
e) Degree of accuracy and finish required

Pattern Allowances

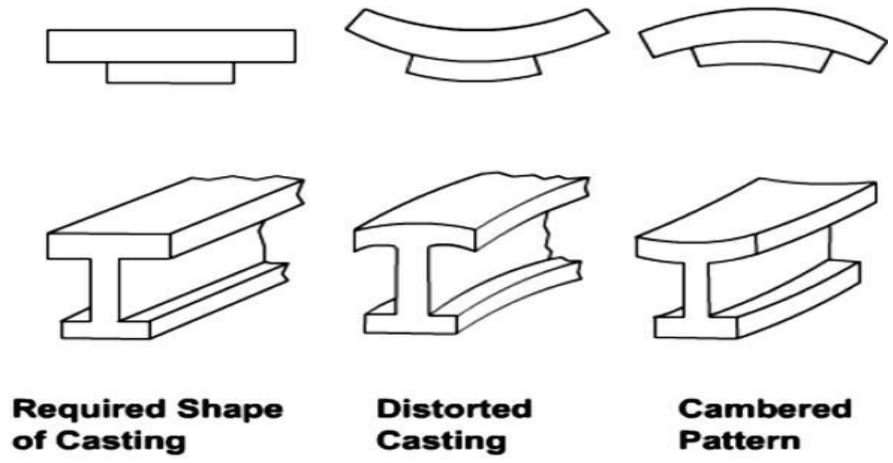
Machining allowances of various metals

Metal	Dimension (inch)	Allowance (inch)
Cast iron	Up to 12	0.12
	12 to 20	0.20
	20 to 40	0.25
Cast steel	Up to 6	0.12
	6 to 20	0.25
	20 to 40	0.30
Non ferrous	Up to 8	0.09
	8 to 12	0.12
	12 to 40	0.16

Pattern Allowances

4. Distortion / Chamber Allowance:

- Due to their typical shapes (U,V,T,L shapes) , castings get distorted during solidification.
- Distortion is observed in irregular castings so that it shrink in uneven manner
- Distortions are caused by internal stresses which are generated on account of unequal cooling of different sections of casting.
- To avoid this distortion allowance is provided
- It varies from 2 to 20 mm



Pattern Allowances

5. Rapping / shake Allowance:

- Before the withdrawal from the sand mould, the pattern is rapped all around the vertical faces to enlarge the mould cavity slightly, which facilitate its removal.
- When pattern is rapped, mould cavity is enlarged.
- To account for this increase, pattern size is reduced.
- This allowance is important in large-sized castings and precision castings.

Amount of rapping allowance depends upon:

a) Extent of rapping
b) Degree of compaction of sand
c) Size of mould
d) Sand type

Types of Moulding Sand

(i) Green Sand:

- When sand is in its natural (more or less moist) state, it is referred to as green sand. It is a mixture of silica sand, with 18 to 30% clay and 6 to 8% water. The clay and water give bonding strength to green sand.
- It is fine, soft, light and porous. Being damp, it retains the shape given to it under pressure during squeezing.
- As the mould becomes dense by ramming, the structure is made porous by venting. Sharp edges are avoided in green sand moulding, because these being weak, break when hot metal is poured.
- Green sand is generally used for casting small or medium sized moulds. Larger output can be obtained from a given floor space as the cost and delay involved in drying the moulds is saved. Coal dust is mixed in green sand to prevent defects in castings.

Types of Moulding Sand

(ii) Dry Sand:

- Dry sand moulding is employed for large castings. The moulds prepared in green sand are dried or baked to remove, almost, all moisture of the moist sand. The structure in the moulding boxes after drying becomes stronger and compact. Venting is therefore necessary but not to that extent, as in the case of green sand mould. For larger heavy moulds, cowdung, horse manure, etc. are mixed with the sand of coarser grains.

Types of Moulding Sand

(iii) Loam Sand:

- It is a mixture of clay and sand milled with water to a thin plastic paste, from which, moulds are built up on a backing of soft bricks.
- Loam sand contains upto 50% clay and dries hard. It also contains fire clay. It must be sufficiently adhesive to hold on to the vertical surfaces of the rough structure of the mould. Chopped stray and manure are commonly used to assist in binding. The moisture content is from 18 to 20%.
- Loam is dried very slowly and completely before it is ready for casting. It is used for casting larger regular shaped castings like chemical pans, drums, etc.

Types of Moulding Sand

(iv) Facing Sand:

- It is used directly next to the surface of the pattern and it comes into contact with the molten metal. Since, it is subjected to the most severe conditions, it must possess high strength and refractoriness. It is made of silica sand and clay, without the addition of used sand.
- Different forms of carbon known as facing materials, (e.g., plumbago powder, ceylon lead or graphite) are used to prevent the metal from burning into the sand. Sometimes they are mixed with 6 to 15 times fine moulding sand to make mould facings.

Types of Moulding Sand

(v) Backing Sand:

- The old, repeatedly used moulding sand, black in colour due to addition of coal dust and burning or coming in contact with molten metal is known as backing sand or floor sand or black sand. It is used to fill in the mould at the back of facing layer. It is weak in bonding strength because the sharp edges of sand grain become rounded due to high temperature of molten metal and burning of clay content.

(vi) System Sand:

- This is used in machine moulding to fill the whole flask. Its strength, permeability, and refractoriness must be higher than those of backing sand.

Types of Moulding Sand

(vii) Parting Sand:

- The moulding boxes are separated from adhering to each other by spreading a fine sharp dry sand called ‘parting sand’. Parting sand is also used to keep the green sand from sticking to the pattern. It is clean clay- free silica sand. Burnt core sand could also be used for this purpose.

(viii) Core Sand:

- It is used for making cores. It is silica sand mixed with core oil (linseed oil, rosin, light mineral oil and other binders). For the sake of economy pitch or flour and water may be used as core sand for large cores.

Types of Moulding Sand

(ix) CO_2-Sand:

- In CO_2 sand, the silica grains, instead of being coated with natural clay, are coated with sodium silicate. This mixture is first packed around the pattern and then hardened by passing CO_2 through the interstices for about a minute. The sand thus sets hard and produces a strong mould.

(x) Shell Sands:

- Shell sands are synthetic sands coated with phenol or urea-formaldehyde resins and cured against a heated pattern to produce very strong, thin shell. No back up sand is required to provide support for the weight of the casting. Since, alloys solidify at high temperatures, the resins are not dissociated. But moulds disintegrate when casting has solidified due to breaking up of chemical bond by heat from solidifying casting.

Properties of Moulding Sand

1. Porosity:

- Porosity also known as permeability is the most important property of the moulding sand. It is the ability of the moulding sand to allow gasses to pass through. Gasses and steam are generated during the pouring of molten metal into the sand cavity. This property depends not only on the shape and size of the particles of the sand but also on the amount of the clay, binding material, and moisture contents in the mixture.

2. Cohesiveness:

- Cohesiveness is the property of sand to hold its particles together. It may be defined as the strength of the moulding sand. This property plays a vital role in retaining intricate shapes of the mould.
- Insufficient strength may lead to a collapse in the mould particles during handling, turning over, or closing. Clay and bentonite improves the cohesiveness.

Properties of Moulding Sand

3. Adhesiveness:

- Adhesiveness is the property of sand due to which the sand particles sticks to the sides of the moulding box. Adhesiveness of sand enables the proper lifting of cope along with the sand.

4. Plasticity:

- Plasticity is the property of the moulding sand by virtue of which it flows to all corners around the mould when rammed, thus not providing any possibility of left out spaces, and acquires a predetermined shape under ramming pressure.

Properties of Moulding Sand

5. Flow-Ability:

- Flow-ability is the ability of moulding sand to free flow and fill the recesses and the fine details in the pattern. It varies with moisture content.

6. Collapsibility:

- Collapsibility is the property of sand due to which the sand mould collapse automatically after the solidification of the casting. The mould should disintegrate into small particles of moulding sand with minimum force after the casting is removed from it.

Properties of Moulding Sand

7. Refractoriness:

- Refractoriness is the property of sand to withstand high temperature of molten metal without fusion or soften.
- Moulding sands with poor refractoriness may burn when the molten metal is poured into the mould. Usually, sand moulds should be able to withstand up to 1650°C.

Special Casting

Extrusion

Extrusion is used for processing most types of metals, thermoplastics and rubbers. The extrusion process is a simple process in which molten metal/polymer is forced through a shaped die using pressure. The pressure is generated from the action of screw rotation against barrel wall.

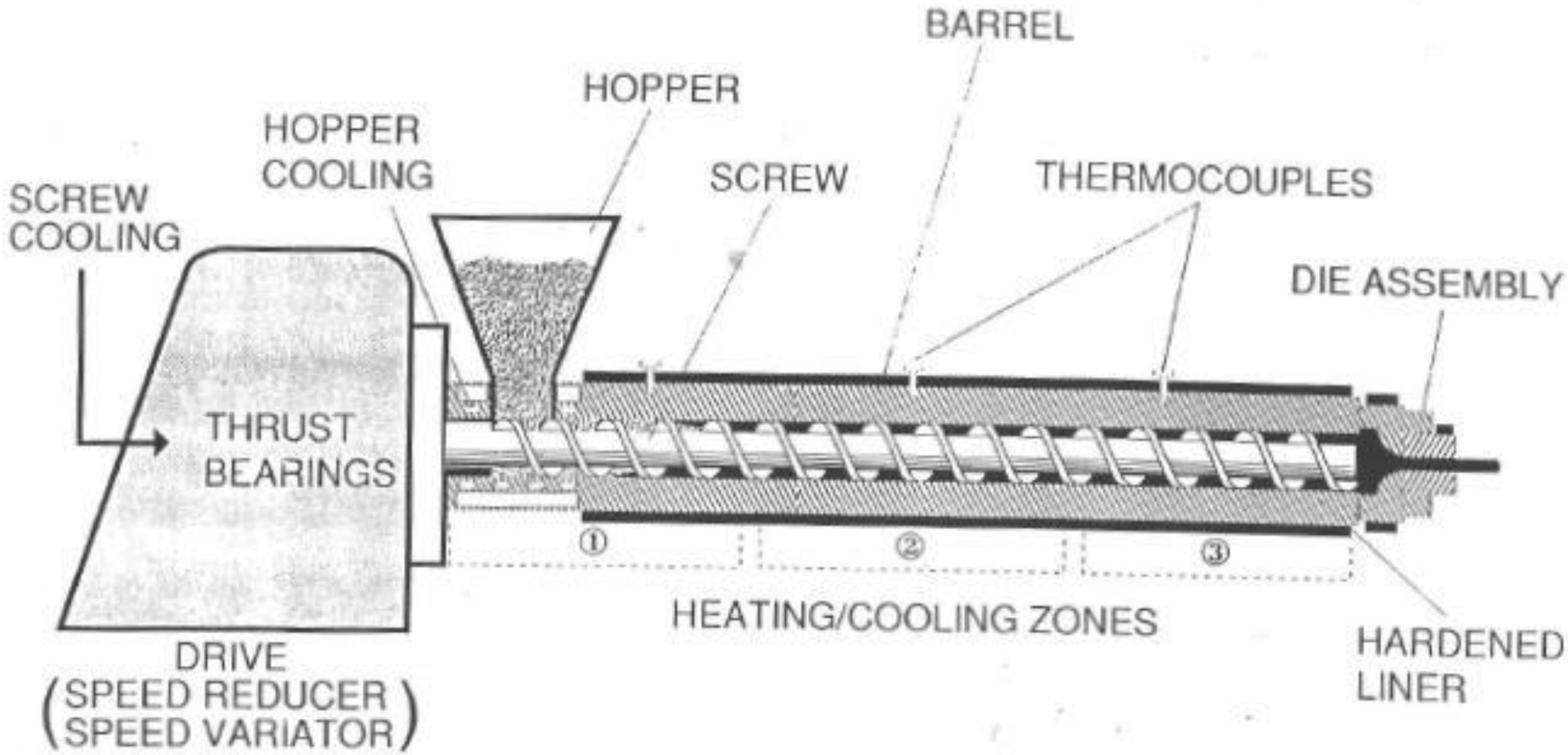
The components of an extruder include:

1. Drive
2. Gearbox and Thrust bearing
3. Feed hopper
4. Barrel

https://www.youtube.com/watch?v=P8BWQBP4Vhk

https://www.youtube.com/watch?v=WaB-dsB1Kfk

BASIC EXTRUDER



Advantages, Disadvantages & Limitations of Extrusion

Affordability

- The plastic extrusion process comes with low costs compared to other moulding processes and greater profit margins, making it an attractive option for businesses that have to process products in bulk.
- Most extrusion moulding uses thermoplastics, which can undergo repeated melting and hardening. Leftover materials, normally discarded as waste in other processes, can be reused, which lowers raw material and disposal costs.

Flexibility

- Extrusion moulding provides considerable flexibility in manufacturing products with a consistent cross-section. So long as the cross-section remains the same, extrusion moulding can produce complex shapes, such as decorative trims.
- With minor alteration to the process, manufacturers use extrusion moulding for plastic sheets. Variation on the extrusion process also allows for the manufacturing of products that mix plastic attributes, such as hard and soft surfaces.

Disadvantages

Size variance: Once the hot plastic is removed from the extruder, it will often expand. The process is called die swell. It is problematic to try and predict the exact degree of expansion. Because it is hard to predict the exact expansion, manufacturers often accept significant levels of deviation from the dimensions of the product.

Limitations

Extrusion plastic moulding does place limits of the types of products that can be manufactured. For example, plastic soda bottles narrow at one end to accommodate a cap, which normal extrusion moulding cannot achieve. Some alternatives require an investment in a different type of extrusion equipment.

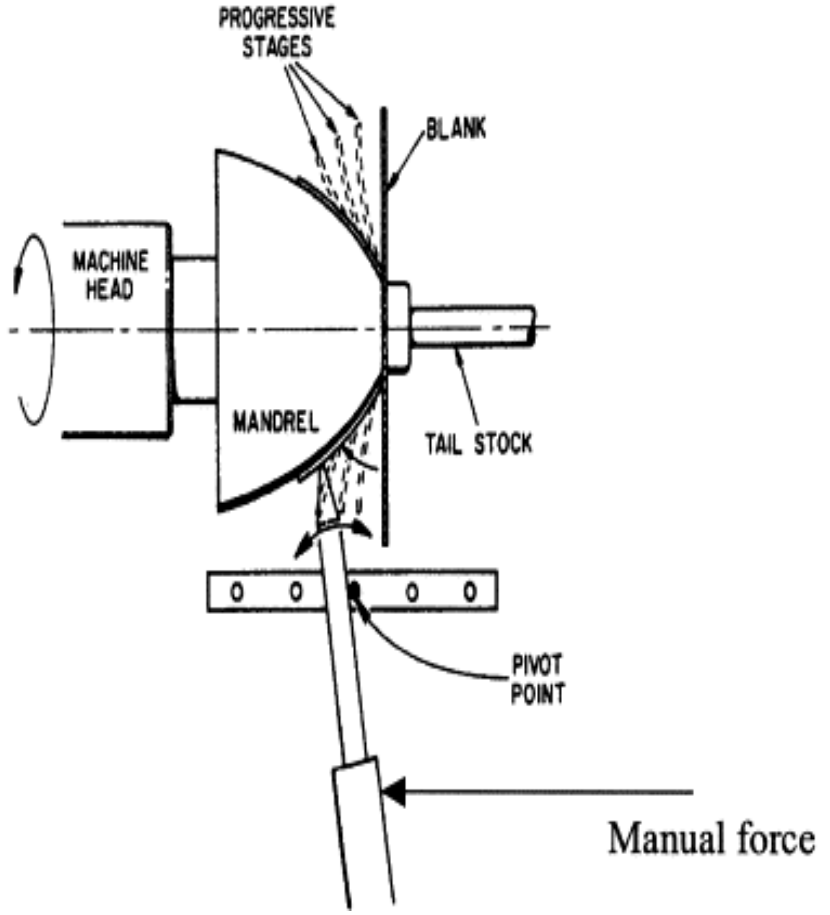
Metal spinning

- **Metal spinning**, also known as **spin forming** or **spinning** or **metal turning** most commonly, is a metalworking process by which a disc or tube of metal is rotated at high speed and formed into an axially symmetric part. Spinning can be performed by hand or by a CNC lathe.
- Metal spinning does not involve removal of material, as in conventional wood or metal turning, but forming (moulding) of sheet material over an existing shape.
- Metal spinning ranges from an artisan's specialty to the most advantageous way to form round metal parts for commercial applications.
- Artisans use the process to produce architectural detail, specialty lighting, decorative household goods and urns. Commercial applications include rocket nose cones, cookware, gas cylinders, brass instrument bells, and public waste receptacles. Virtually any ductile metal may be formed, from aluminium or stainless steel, to high-strength, high-temperature alloys including INX, Inconel, Grade 50 / Corten, and Hastelloy. The diameter and depth of formed parts are limited only by the size of the equipment available.

https://www.youtube.com/watch?v=43N44ICyuEU

Spinning process

The spinning process is fairly simple. A formed block is mounted in the drive section of a lathe. A pre-sized metal disk is then clamped against the block by a pressure pad, which is attached to the tailstock.
The block and workpiece are then rotated together at high speeds. A localized force is then applied to the workpiece to cause it to flow over the block. The force is usually applied via various levered tools.
Simple workpieces are just removed from the block, but more complex shapes may require a multi-piece block. Extremely complex shapes can be spun over ice forms, which then melt away after spinning. Because the final diameter of the workpiece is always less than the starting diameter, the workpiece must thicken, elongate radially, or buckle circumferentially.



Hydro forming process

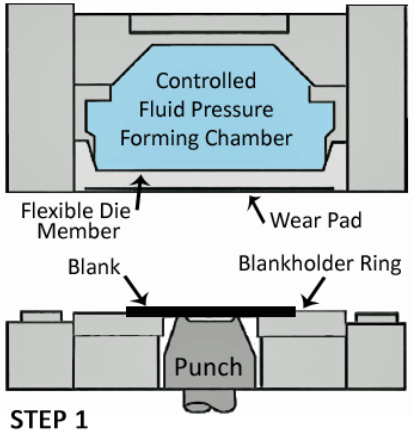
- **Hydroforming** is a cost-effective way of shaping ductile metals such as aluminium, brass, low alloy steel, and stainless steel into lightweight, structurally stiff and strong pieces.
- One of the largest applications of hydroforming is the automotive industry, which makes use of the complex shapes made possible by hydroforming to produce stronger, lighter, and more rigid unibody structures for vehicles.
- This technique is particularly popular with the high-end sports car industry and is also frequently employed in the shaping of aluminium tubes for bicycle frames.
- Hydroforming is a specialized type of die forming that uses a high pressure hydraulic fluid to press room temperature working material into a die.

https://www.youtube.com/watch?v=1QPSCXEoJKQ

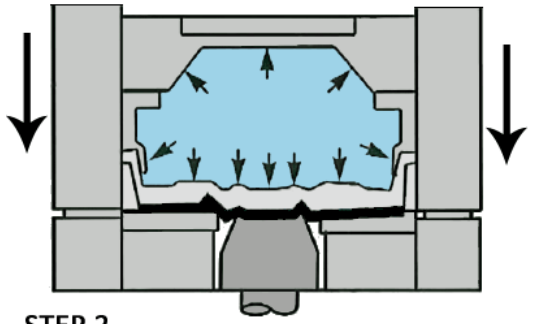
Hydro forming process

- To hydroform aluminium into a vehicle's frame rail, a hollow tube of aluminium is placed inside a negative mold that has the shape of the desired result.
- High pressure hydraulic pumps then inject fluid at very high pressure inside the aluminium tube which causes it to expand until it matches the mold.
- The hydroformed aluminium is then removed from the mold.
- Hydroforming allows complex shapes with concavities to be formed, which would be difficult or impossible with standard solid die stamping. Hydroformed parts can often be made with a higher stiffness-to-weight ratio and at a lower per unit cost than traditional stamped or stamped and welded parts.
- Virtually all metals capable of cold forming can be hydroformed, including aluminium, brass, carbon and stainless steel, copper, and high strength alloys.

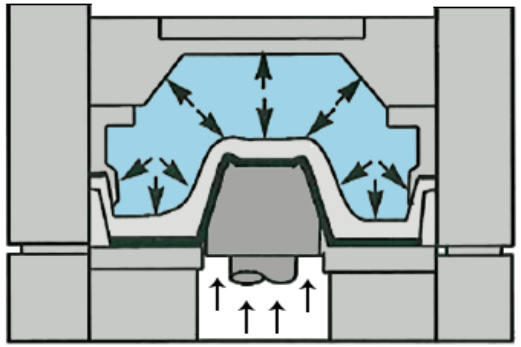
Hydro forming process



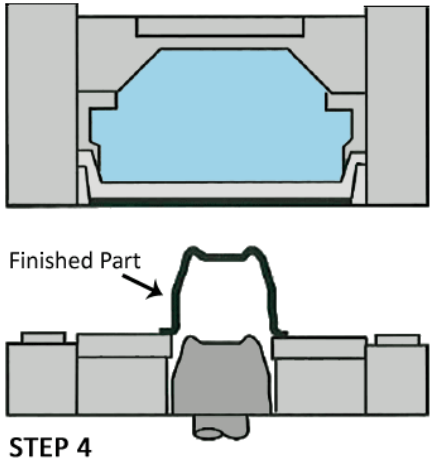
STEP 1
A metal blank is placed on blankholder ring.



STEP 2
Top of press lowered and fluid chamber pressurized.



STEP 3
Punch pushes upward, forming metal blank against pressure chamber.



STEP 4
Top of press raised, pressure released, and punch lowered. The part is now finished.

A hydroforming press operates like the upper or female die element. This consists of a pressurized forming chamber of oil, a rubber diaphragm and a wear pad. The lower or male die element, is replaced by a punch and ring. The punch is attached to a hydraulic piston, and the blank holder, or ring, which surrounds the punch.

The hydroforming process begins by placing a metal blank on the ring. The press is closed bringing the chamber of oil down on top of the blank. The forming chamber is pressurized with oil while the punch is raised through the ring and into the forming chamber. Since the female portion of this forming method is rubber, the blank is formed without the scratches associated with stamping.

The diaphragm supports the entire surface of the blank. It forms the blank around the rising punch, and the blank takes on the shape of the punch. When the hydroforming cycle is complete, the pressure in the forming chamber is released and the punch is retracted from the finished part.

Advantages of hydro forming process

- Freedom of design for complex parts
- Reduced number of individual parts of a body in white
- Minimization of assembled and/or welded connections
- Reduction of material and weight of part
- Higher mechanical strength
- Longer durability
- High dimensional and forming accuracy with reduced springback

DISADVANTAGES

Hydro Forming also has some drawbacks,

- Slow cycle time.
- Expensive equipment.
- Lack of extensive knowledge base for process and tool design

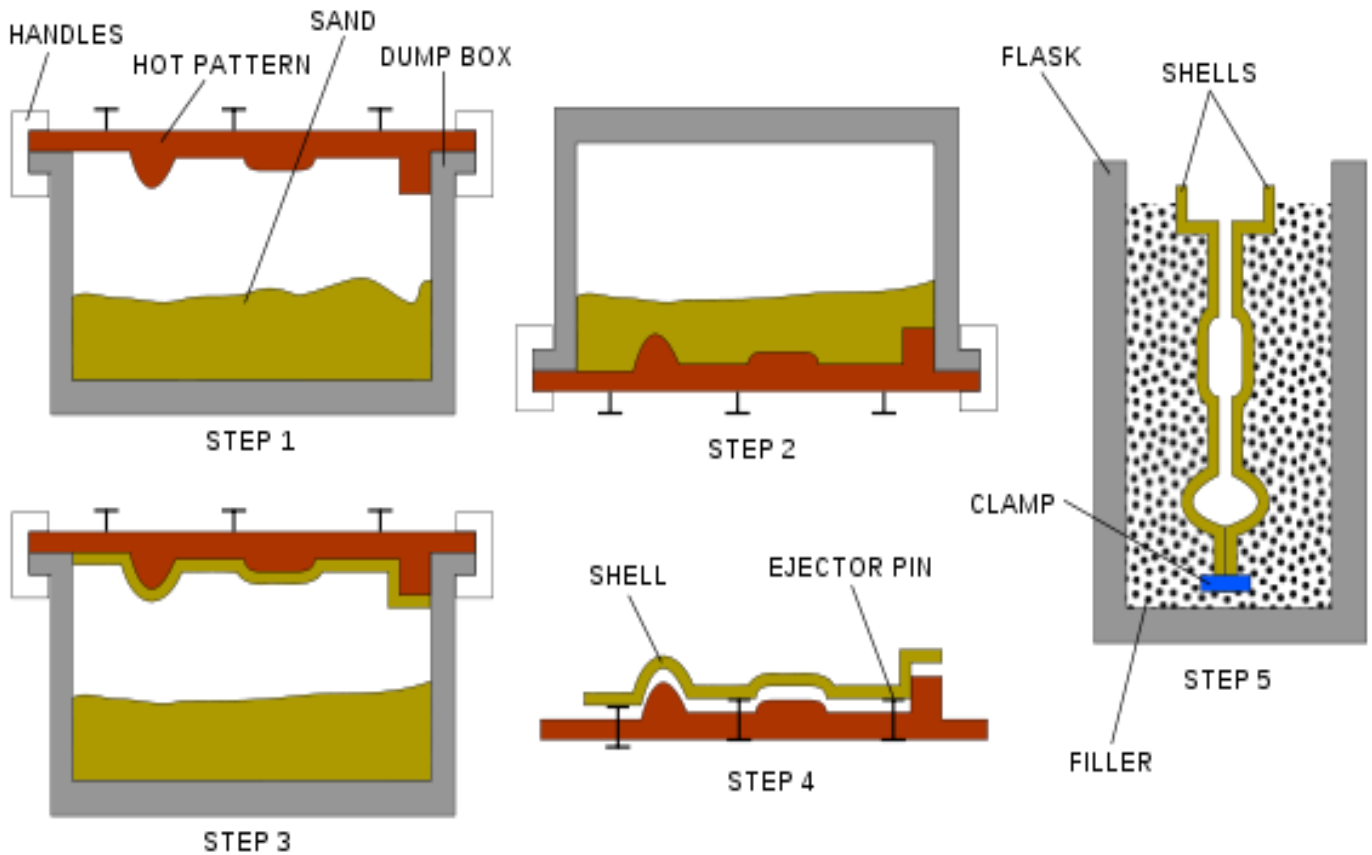
Shell Mold Casting

- Shell mold casting or shell molding is a metal casting process in manufacturing industry in which the mold is a thin hardened shell of sand and thermosetting resin binder, backed up by some other material. Shell molding was developed as a manufacturing process during the mid-20th century in Germany. Shell mold casting is particularly suitable for steel castings under 20 lbs; however almost any metal that can be cast in sand can be cast with the shell molding process. Also much larger parts have been manufactured with shell molding. Typical parts manufactured in industry using the shell mold casting process include cylinder heads, gears, bushings, connecting rods, camshafts and valve bodies.

Shell Mold Process

- The first step in the shell mold casting process is to manufacture the shell mold. The sand we use for the shell molding process is of a much smaller grain size than the typical green sand mold. This fine grained sand is mixed with a thermosetting resin binder. A special metal pattern is coated with a parting agent, (typically silicone), which will latter facilitate in the removal of the shell. The metal pattern is then heated to a temperature of 350F-700F degrees, (175C-370C).

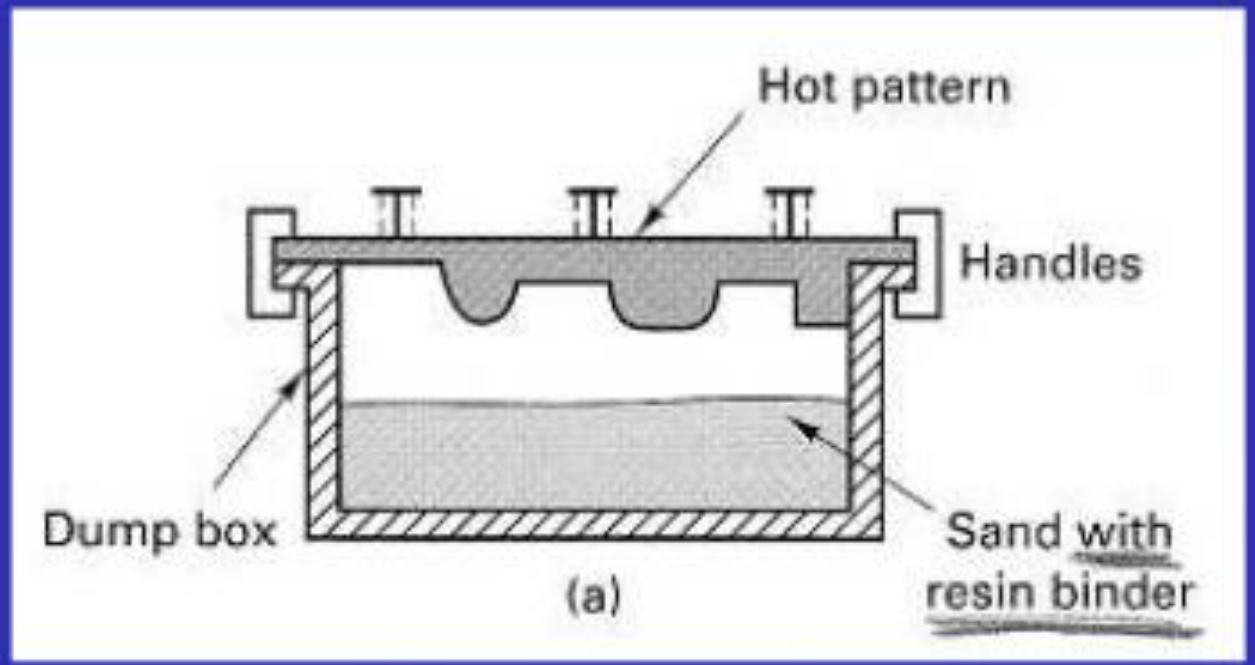
Shell Mold Process



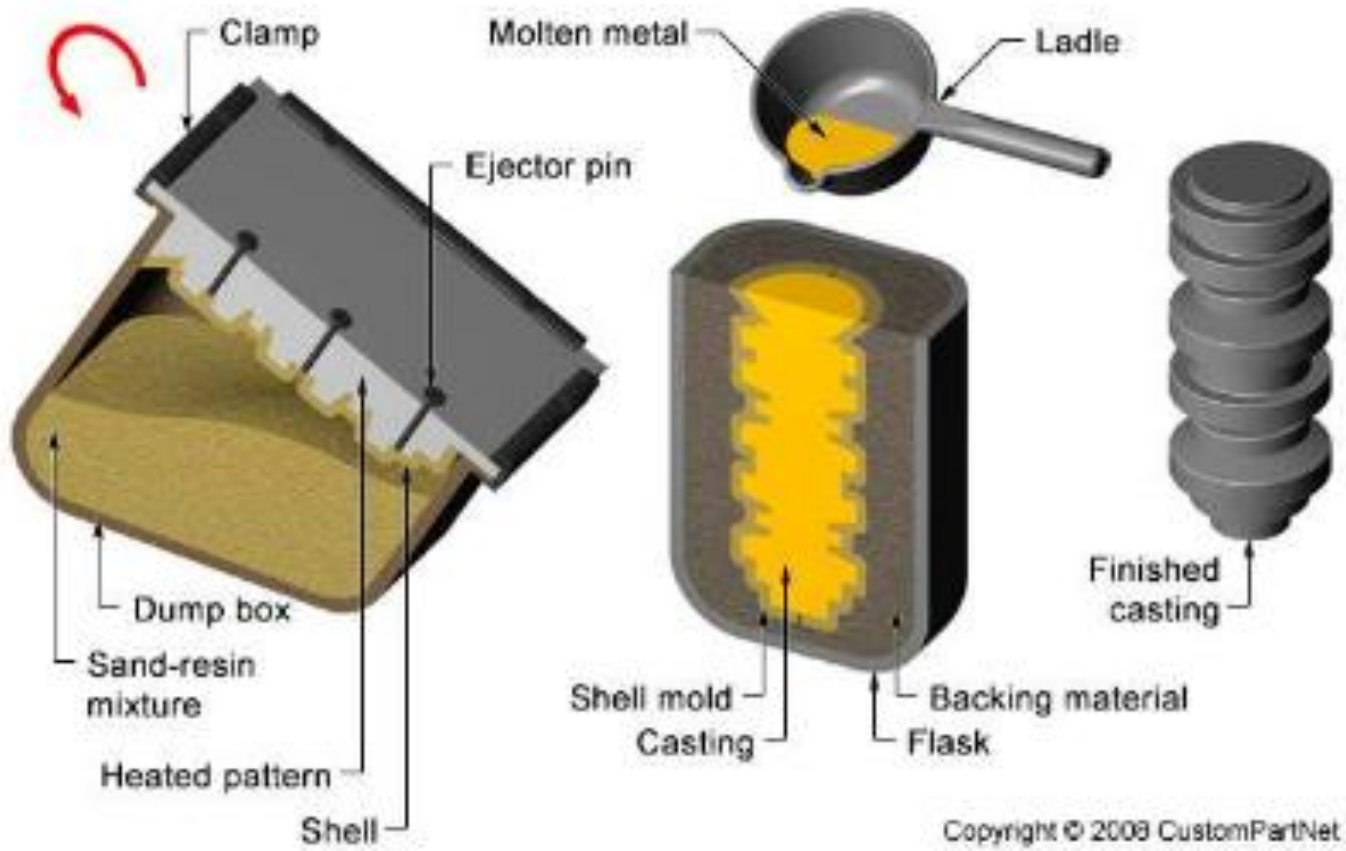
SHELL-MOLD CASTING

(Sequential Operations)

(a) A heated pattern is placed over a dump box containing a sand and resin mixture.



Shell Mold Process



Advantages:

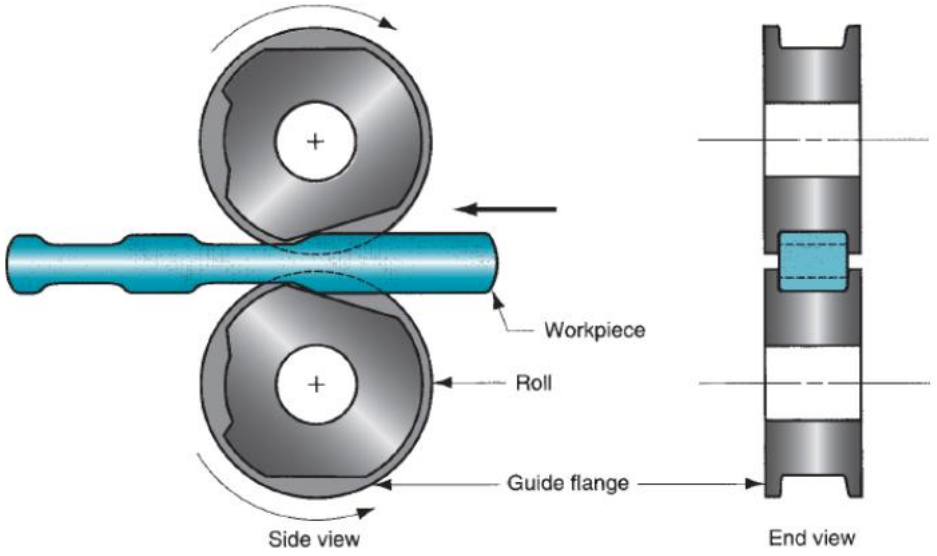
- Can form complex shapes and fine details,
- Very good surface finish,
- High production rate, Low labor cost,
- Low tooling cost, Little scrap generated.
- Can produce very large parts,
- Can form complex shapes,
- Many material options,
- Low tooling and equipment cost,

Disadvantages:

- High equipment cost
- Poor material strength
- High porosity possible
- Poor surface finish and tolerance
- Secondary machining often required.

Rolling

- Roll forging: It is a forming process used to reduce the cross section of a cylindrical or rectangular rod by passing it through a set of opposing rolls that have matching grooves w.r.t. the desired shape of the final part. It combines both rolling and forging, but classified as forging operation. Depending on the amount of deformation, the rolls rotate partially. Roll-forged parts are generally stronger and possess desired grain structure compared to machining that might be used to produce the same part.



Forging

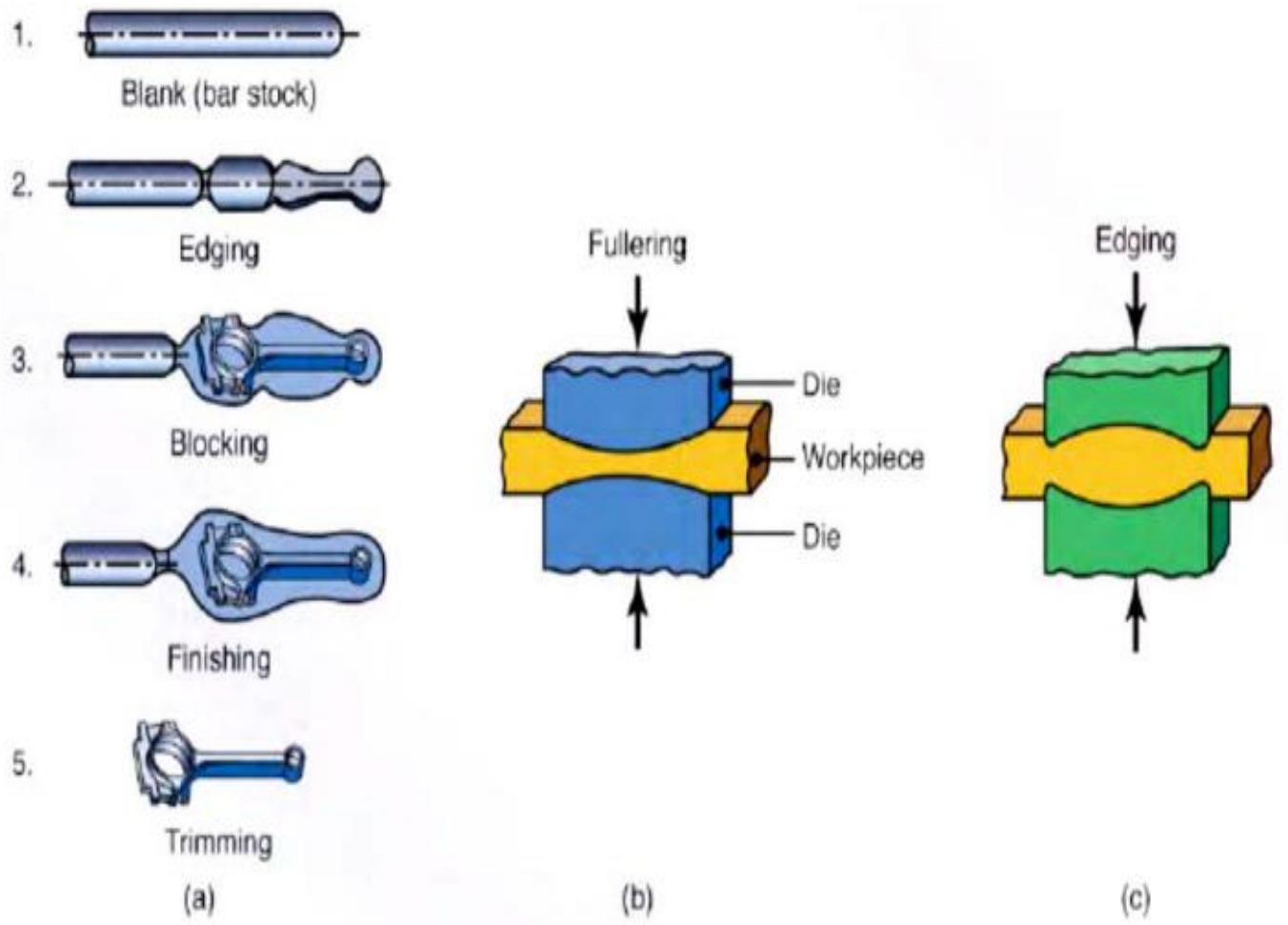
- Forging is the working of metal into a useful shape by hammering or pressing.
- The oldest of the metalworking arts (primitive blacksmith). • Replacement of machinery occurred during early the Industrial revolution.
- Forging machines are now capable of making parts ranging in size of a bolt to a turbine rotor.
- Most forging operations are carried out hot, although certain metals may be cold forged

Forging

- Forging is perhaps oldest metal working process and was known even during prehistoric days when metallic tools were made by heating and hammering.
- Forging is basically involves plastic deformation of material between two dies to achieve desired configuration.
- Depending upon complexity of the part forging is carried out as open die forging and closed die forging.
- In open die forging, the metal is compressed by repeated blows by a mechanical hammer and shape is manipulated manually.
- In closed die forging, the desired configuration is obtained by squeezing the workpiece between two shaped and closed dies.

Stages in forging a connecting rod for an internal combustion engine

a) bar stock;
b) fullering,
c) edging
d) rolling,
e) blocking,
f) finishing,
g) trimming;



Smith forging

- Smith forging is also called flat die and open die forging. It includes the broad field for forging work produced between flat faced sizes and possibly supplemented by stock tooling.
- The final shape of the forging depends on the skill of the smith fir size and shape.
- It produces work pieces of lesser accuracy as compared to impression or closed die forging.
- Tooling is simple, inexpensive and allows the production of a large variety of shapes.

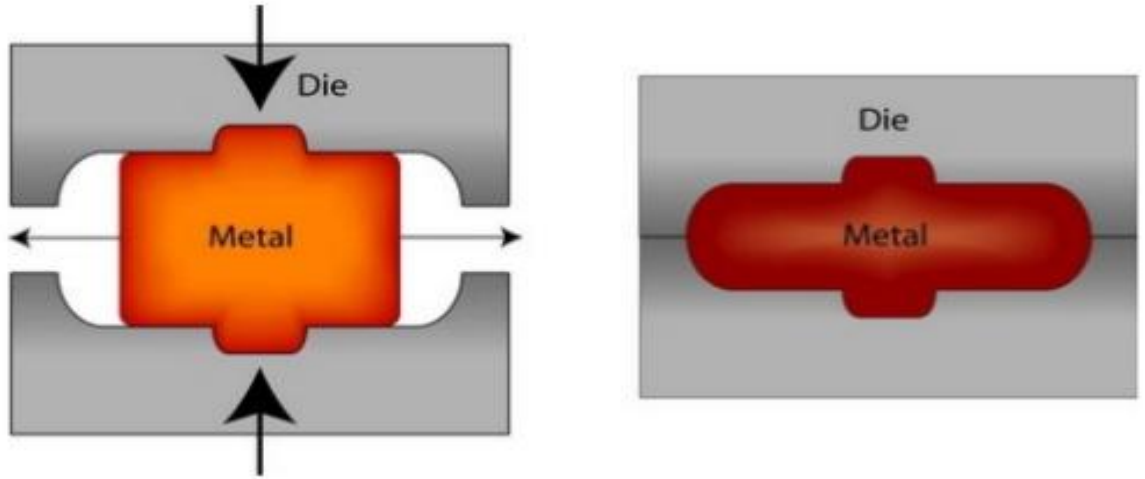
Smith Forging





Drop forging

- Drop forging sometimes referred to as impression die forging comprises of die on the anvil which resembles a mould, the ram which falls and strikes the top of the workpiece can also be equipped with die. The metal work piece is heated and placed on the lower die while the ram falls down forcing the metal to fill the contours of the die blocks



Press forging

- Press forging is done in presses. The action is relatively slow squeezing instead of delivering heavy blows and penetrates deeply because it gives the metal time to flow
- Press forgings are shaped at each impression with a single smooth stroke and they stick to the die impression more rigidly

